

EXECUTIVE SUMMARY

Project: Solving Inventory Inefficiencies Using SQL

Team Name: Lazy Peeps

Tools Used: MySQL, Jupyter Notebook, Power BI

PROJECT OVERVIEW :

This project aims to simulate the responsibilities of a data analyst working in a retail environment. Our goal was to build an SQL-driven inventory monitoring system which can:

- Track stock levels
- Detect low inventory conditions
- Estimate reorder points using historical sales
- Analyze inventory turnover
- Support smarter, data-backed decisions using KPIs

We started by designing a clean database schema, moving on to writing scalable SQL scripts, and preparing actionable outputs that were further visualized in a Power BI dashboard.

KEY INSIGHTS :

- *Low Inventory Problem*
Over 70 product-store combinations were found with low inventory levels. These products are at risk of stockouts and should be prioritized for replenishment.
- *Inconsistent Demand Trends*
Several products showed a sharp drop in average sales month-over-month, indicating demand uncertainty or overstocking in prior cycles.
- *Slow vs Fast-Moving Products*
Products were categorized based on inventory turnover ratios:
 - Fast-moving: High inventory turnover ratio
 - OK inventory flow : Moderate inventory turnover ratio
 - Slow-moving: Low inventory turnover ratio
- *Best-Selling Products by Region*
Regional analysis showed different top sellers in each zone — this is valuable for location-based stocking.
- *Days of Inventory Left*
Some products may run out within a week; others are heavily overstocked — highlighting urgent and excess stock.

- *High Revenue Contributors*

A select few SKUs generate a large portion of revenue. These should be the focus of future promotions and pricing decisions.

BUSINESS RECOMMENDATIONS :

- ✓ Restocking should be automated using reorder points
- ✓ Inventory holding costs can be reduced via turnover analysis
- ✓ Stock should be customized by regional sales data
- ✓ Use dashboard alerts to avoid future stockouts
- ✓ Slow moving SKUs should be promoted .
- ✓ Regular analysis should be done to observe the changes in subsequent months

This project has given us a real-world view into how data analytics drives retail efficiency. With SQL-backed insights and dynamic dashboards, we moved beyond numbers — toward smarter and data-informed action.

Thank You !!